Exercise 1

We just compute the truth tables of the propositions

a)

P	Q	$P \vee Q$	$P \wedge Q$	$\neg(P \wedge Q)$	$P \vee Q \Leftrightarrow \neg(P \wedge Q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

The first equivalence is false.

b)

P	Q	$P \Rightarrow Q$	$\neg(P \Rightarrow Q)$	$\neg Q$	$P \wedge \neg Q$	$\neg(P \Rightarrow Q) \Leftrightarrow P \wedge \neg Q$
T	T	T	F	F	F	T
T	F	F	T	T	T	T
F	T	T	F	F	F	T
F	F	T	F	T	F	T

This second equivalence is true

Exercise 2

a) $\exists\, n \in \mathbb{N}$ s.t. n is odd

b) $\exists\, \varepsilon > 0$ s.t. $\forall\, x_0 > 0\ \exists\, x > x_0$ s.t. $\left|\frac{1}{x}\right| > \varepsilon$

c) FALSE: Given any $a \in \mathbb{R}$ we can pick $b = a^2 + 1$ obtaining $a^2 \geq a^2 + 1$ which is false

d) TRUE: $\forall\, b \in \mathbb{R}$ we can pick $a = \sqrt{|b|}$ obtaining $|b| \geq b$ which is always true

Exercise 3

a) We need to prove that $\forall x \in G \quad x \in (A \cap B)^c \Leftrightarrow x \in A^c \cup B^c$

$$
\begin{aligned}
x \in (A \cap B)^c &\Leftrightarrow \neg(x \in A \cap B) \\
&\Leftrightarrow \neg(\underbrace{x \in A}_{P} \wedge \underbrace{x \in B}_{Q}) \\
&\Leftrightarrow \neg(P \wedge Q) \\
&\Leftrightarrow \neg P \vee \neg Q \\
&\Leftrightarrow \neg(x \in A) \vee \neg(x \in B) \\
&\Leftrightarrow x \in A^c \vee x \in B^c \\
&\Leftrightarrow x \in A^c \vee B^c
\end{aligned}
$$

This can be checked with a truth table

b) We need to prove that $\forall x \in G \quad x \in (A_1 \cap A_2) \cup B \Leftrightarrow x \in (A_1 \cup B) \cap (A_2 \cup B)$

$$
\begin{aligned}
x \in (A_1 \cap A_2) \cup B &\Leftrightarrow (x \in A_1 \cap A_2) \vee x \in B \\
&\Leftrightarrow (\underbrace{x \in A_1}_{P} \wedge \underbrace{x \in A_2}_{Q}) \vee \underbrace{x \in B}_{R} \\
&\Leftrightarrow (P \wedge Q) \vee R \\
&\Leftrightarrow (P \vee R) \wedge (Q \wedge R) \\
&\Leftrightarrow (x \in A_1 \vee x \in B) \wedge (x \in A_2 \vee x \in B) \\
&\Leftrightarrow (x \in A_1 \cup B) \wedge (x \in A_2 \cup B) \\
&\Leftrightarrow x \in (A_1 \cup B) \cap (A_2 \cup B).
\end{aligned}
$$

Check with a truth table

Exercise 4

a) NOT INJECTIVE: since $\sin(0) = \sin(2\pi)$

NOT SURJECTIVE: since $\sin(x) = 2$ has no solution in $\mathbb{R}$.

b) i) NOT INJECTIVE: $f(-1) = 0 = f(1)$

SURJECTIVE: For $y \in [0,1]$ $x_y = \sqrt{1-y^2} \in [0,1] \subset [-1,1] = A$ is a solution for $f(x_y) = y$.

In fact $f(x_y) = \sqrt{1-(\sqrt{1-y^2})^2} = \sqrt{1-1+y^2} = y$ since $y \geq 0$.

ii) INJECTIVE: Suppose $f(x) = f(x')$, then $\sqrt{1-x^2} = \sqrt{1-x'^2}$

Since both sides are ≥ 0 we can take the square,

$1-x^2 = 1-x'^2 \Rightarrow x^2 = x'^2 \Rightarrow x = x'$ since they are ≥ 0

SURJECTIVE: Same proof as above.

c) INJECTIVE: Suppose $f(n) = f(m)$, then $(n-\frac{1}{3})^2 = (m-\frac{1}{3})^2$

then $n-\frac{1}{3} = m-\frac{1}{3}$ or $n-\frac{1}{3} = -m+\frac{1}{3}$

In the first case $n=m$ as whished.

In the second case we get $n = -m+\frac{2}{3}$

which is impossible since n and m are integers.

NOT SURJECTIVE: $f(n) \geq 0$ hence $f(n) = -1$ has no solutions

d) INJECTIVE: Suppose $f(m,n) = f(m',n')$.

Set $s = m+n$ and $s' = m'+n'$, then we have

$\frac{s(s+1)}{2} + m = \frac{s'(s'+1)}{2} + m' \Rightarrow \frac{s(s+1)}{2} - \frac{s'(s'+1)}{2} = m'-m$.

If $s=s'$ we get $m'=m$ and consequently $(m,n)=(m',n')$.

Let's suppose by contradiction that $s>s'$, we have

$\frac{s(s+1)}{2} - \frac{s'(s'+1)}{2} \geq \frac{(s'+1)(s'+2)}{2} - \frac{s'(s'+1)}{2} \geq s'+1$

$> s \geq m' \geq m'-m$ contradiction

SURJECTIVE: Fix $k \in \mathbb{N}$. If we let s vary in $\mathbb{N}$ $\frac{s(s+1)}{2}$ is an increasing sequence of natural numbers with 0 as first term. Therefore $\exists!\ s \in \mathbb{N}$ s.t.

$\frac{s(s+1)}{2} \leq k < \frac{(s+1)(s+2)}{2}$

Notice that $k - \frac{s(s+1)}{2} < \frac{(s+1)(s+2)}{2} - \frac{s(s+1)}{2} = s+1$

Let's set $m = k - \frac{s(s+1)}{2}$ and $s = m+n$

Notice that $0 \leq m \leq s$ and $n \geq 0$

One can check that $f(m,n) = \frac{(m+n)(m+n+1)}{2} + m = \frac{s(s+1)}{2} + m = k$

Exercise 5

a) We want to prove that $x \in A \Rightarrow x \in \epsilon^{-1}(\epsilon(A))$

Suppose $x \in A$, then by definition $\epsilon(x) \in \epsilon(A)$.

Recall that $\epsilon^{-1}(\epsilon(A)) = \{x' : \epsilon(x') \in \epsilon(A)\}$

Hence $x \in \epsilon^{-1}(\epsilon(A))$.

b) Under the additional hypotesis of ϵ being injective we want to prove that $x \in \epsilon^{-1}(\epsilon(A)) \Rightarrow x \in A$.

$x \in \epsilon^{-1}(\epsilon(A))$ implies that $\epsilon(x) \in \epsilon(A)$, hence, by definition $\exists\, x' \in A$ s.t. $\epsilon(x) = \epsilon(x')$. Since ϵ is injective $x = x'$ and $x \in A$.

c) Suppose that $\epsilon(x) = \epsilon(x')$

Then $x, x' \in \epsilon^{-1}(\epsilon(\{x\}))$. Let's pick $A = \{x\}$, then $x' \in \epsilon^{-1}(\epsilon(\{x\})) = \{x\} \Rightarrow x' = x$ as desired.